
Book Store Management System

SQL Project Report

Cover Page

Include:

- **Project Title:** Book Store Management System
 - **Technology:** PostgreSQL
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 - **Role:** Aspiring Data Analyst
 - **Date:** July 2026
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1. Project Overview

This project demonstrates how SQL can be used to manage and analyze bookstore operations. It stores information about books, customers, orders and inventory to generate meaningful business insights.

2. Business Problem

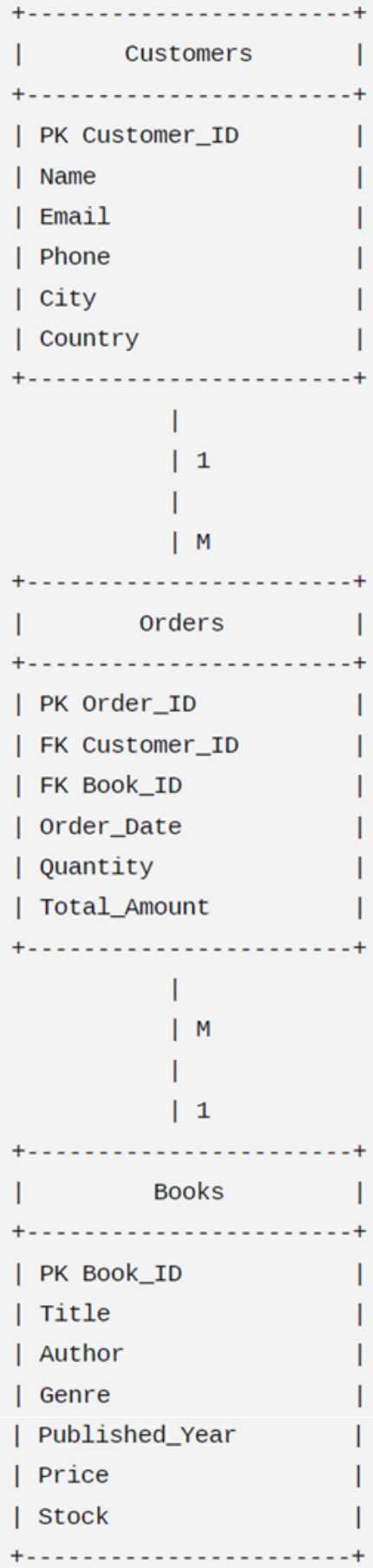
- Track customer orders
 - Manage inventory
 - Monitor book sales
 - Identify top-selling books
 - Analyze customer purchasing behavior
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3. Project Objectives

- Design a relational database
 - Create normalized tables
 - Establish relationships using primary and foreign keys
 - Perform sales analysis
 - Generate business reports using SQL
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4. ER Diagram and Database Schema

**ER- Diagram and
Database Schema**



5. SQL Concepts Used

- CREATE TABLE
 - INSERT
 - PRIMARY KEY
 - FOREIGN KEY
 - JOIN
 - LEFT JOIN
 - GROUP BY
 - HAVING
 - Aggregate Functions
 - COALESCE Function
-

6. Business Questions

1. Retrieve the total stock of books available?
 2. Find the details of the most expensive book?
 3. List all genres available in the Books table?
 4. Find the book with the lowest stock?
 5. Calculate the total revenue generated from all orders?
 6. Retrieve the total number of books sold for each genre?
 7. Find the most frequently ordered book?
 8. Retrieve the total quantity of books sold by each author?
 9. Find the customer who spent the most on orders?
 10. Calculate the stock remaining after fulfilling all orders?
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7. Business Insights

- Mystery books generated the highest revenue.
- Kim Turner is the customer who spent most on orders.
- “Devolved zero administration process improvement” is the most frequently ordered book.
- The total revenue generated from all orders is \$75629.

8.Challenges Faced

- Designing relationships between multiple tables
 - Maintaining referential integrity
 - Writing complex JOIN queries
 - Optimizing SQL queries
 - Avoiding duplicate records
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9.Learning Outcomes

- Improved SQL query writing
 - Learned database normalization
 - Gained experience with JOINS and aggregations
 - Converted business questions into SQL queries
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10. Future Enhancements

- Connect the database to Power BI
 - Build an interactive orders dashboard
 - Automate reports using Python
 - Add customer recommendation features
 - Develop sales forecasting models
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11. Conclusion

The Book Store Management System project demonstrates practical SQL skills for designing relational databases, writing analytical queries, and generating business insights. It showcases the ability to solve real-world business problems using PostgreSQL.